Problem 6

Numerically implement the CG(3) method on a uniform mesh with the following functions

$$\begin{aligned}
f(x) &= p_1(x)\cos(2\pi x) + p_2(x)\sin(2\pi x),\\
p_1(x) &= -64\pi(18 - 6\pi^4x^2 + 18x^3\pi^2 + 2\pi^4x - 36\pi^2x^2 - 27x + 12\pi^2x + 4\pi^4x^3 + 3\pi^2),\\
p_2(x) &= 216 - 816\pi^2 - 96\pi^4 + 32\pi^6x^2 - 384\pi^4x^3 - 64\pi^6x^3 - 288\pi^4x^2\\
&\quad + 3456\pi^2x + 576\pi^4x - 2592\pi^2x^2 + 144\pi^4x^4 + 32\pi^6x^4,\\
u(x) &= \big((18 + 2\pi^2)x^2 + (-24 - 4\pi^2)x^3 + (9 + 2\pi^2)x^4\big)\sin(2\pi x). \qquad (6.1)
\end{aligned}$$

Numerically determine the convergence rate in both the energy norm and the L^2-norm by creating an appropriate convergence table.

Step 1 — the data is consistent

- Checked symbolically before computing anything:

$$u^{(iv)} - f \equiv 0, \qquad u(0) = 0, \quad u'(0) = 0, \quad u''(1) = 0, \quad u'''(1) = 0.$$

- So (6.1) is a manufactured solution for exactly the cantilever of Problems 1–5, and u is available to measure the error against.

Step 2 — the implementation

- **Mesh.** N_e uniform elements, $h = 1/N_e$, nodes $x_i = ih$. In the lecture's labelling $M = N_e - 1$ interior nodes.
- **Unknowns.** Two per node. Node 0 is clamped, so both of its DOFs are deleted; the free tip keeps both. $\dim V_h = 2N_e$.
- **Stiffness.** Assembled element by element from the Hermite matrix with $EI = 1$ (Problem 3),

$$\boldsymbol{k}^e = \frac{1}{h^3}\begin{bmatrix} 12 & 6h & -12 & 6h \\ 6h & 4h^2 & -6h & 2h^2 \\ -12 & -6h & 12 & -6h \\ 6h & 2h^2 & -6h & 4h^2 \end{bmatrix},$$

  which reproduces the $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}$ blocks of Problem 3 exactly (checked against them directly).
- **Load.** $b_i = \int_0^1 f\phi_i \,\mathrm{d}x$ evaluated by 12-point Gauss–Legendre quadrature per element — *not* by interpolating f. f in (6.1) is a polynomial of degree ≤ 4 times $\sin$ / $\cos(2\pi x)$, so the rule is far more than enough; the results are identical to 9 digits with 8, 16 or 24 points.
- **Solve.** $\boldsymbol{S}\boldsymbol{d} = \boldsymbol{b}$ on the free DOFs.
- **Errors.** On each element, by 20-point Gauss,

$$\|e\|_E = \Big(\int_0^1 \big(u'' - U''\big)^2 \,\mathrm{d}x\Big)^{1/2}, \qquad \|e\|_{L^2} = \Big(\int_0^1 \big(u - U\big)^2 \,\mathrm{d}x\Big)^{1/2},$$

  with u'' differentiated in closed form (checked against a symbolic derivative to 5×10^{-13}).
- **Rate.** Successive meshes halve h, so rate $= \log_2\big(E(h)/E(h/2)\big)$.

Step 3 — convergence table

elements	M	h	$\|e\|_E$	rate	$\|e\|_{L^2}$	rate
4	3	0.25000	2.3314×10^{1}	—	6.4465×10^{-2}	—
8	7	0.12500	5.6259×10^{0}	2.05	3.9018×10^{-3}	4.05
16	15	0.06250	1.4162×10^{0}	1.99	2.4618×10^{-4}	3.99
32	31	0.03125	3.5490×10^{-1}	2.00	1.5434×10^{-5}	4.00
64	63	0.01562	8.8780×10^{-2}	2.00	9.6542×10^{-7}	4.00
128	127	0.00781	2.2199×10^{-2}	2.00	6.0369×10^{-8}	4.00

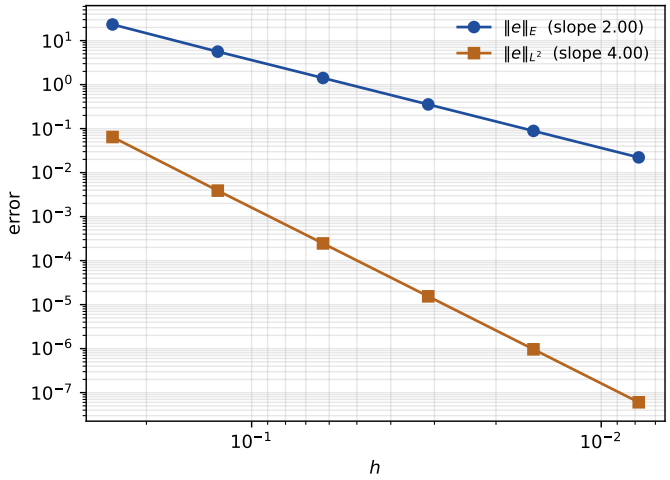


Figure 1: Convergence of the CG(3) cantilever solution under uniform refinement. Energy norm $O(h^2)$, L^2 norm $O(h^4)$.

Step 4 — the measured rates

- **Energy norm: rate $\to 2.00$.** This is the prediction of Problem 4, $\|e\|_E \le c_3 h^2 \|u^{(4)}\| = \mathcal{O}(h^2)$.
- **L^2 norm: rate $\to 4.00$.** This is the prediction of Problem 5, $\|e\|_{L^2} \le c_3^2 h^4 \|u^{(4)}\| = \mathcal{O}(h^4)$.
- The coarsest pair of meshes shows 2.05 and 4.05 — the asymptotic rate is not yet reached at $h = 1/4$, where the mesh barely resolves $\sin(2\pi x)$. From $h = 1/16$ down, both rates are clean to two decimals.
- Both estimates are therefore **sharp**: the observed rates are not merely $\ge$ the predicted ones, they equal them.

Conclusion. CG(3) on the cantilever converges at $\mathcal{O}(h^2)$ in the energy norm and $\mathcal{O}(h^4)$ in the L^2 norm, confirming Problems 4 and 5. The displacement converges two orders faster than the curvature, i.e. than the bending moment.